Question	Answer	Mark
4(a)	(Young modulus =) stress / strain	**B1**
4(b)(i)	$k = F / \Delta L$ **or** 1 / gradient	**C1**
	$= 90 \times 10^3 / (2 \times 10^{-3})$ (*or other point on line*) $= 4.5 \times 10^7\,\text{N m}^{-1}$	**A1**
4(b)(ii)	$E = ½F\Delta L$ **or** $E = ½k(\Delta L)^2$	**C1**
	$= ½ \times 90 \times 10^3 \times 2 \times 10^{-3}$ **or** $½ \times 4.5 \times 10^7 \times (2 \times 10^{-3})^2$	**C1**
	$= 90\,\text{J}$	**A1**
4(c)	straight line starting from (0, 150) and below original line	**M1**
	line ends at (90, 147)	**A1**

Question	Answer	Marks
5(a)	intensity ∝ (amplitude)^2	B1